

12.4

Disk Scheduling

(a) FCFS

- The simplest form of disk scheduling.
- Generally does not provide the fastest service.

Example 3 :

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We illustrate the **FCFS** disk scheduling algorithms with the following information to get the total distance (in cylinders) :

- ✓ A disk drive with 200 cylinders (0-199);
- ✓ A request queue :

  98, 183, 37, 122, 14, 124, 65, 67
- ✓ Head pointer 53.



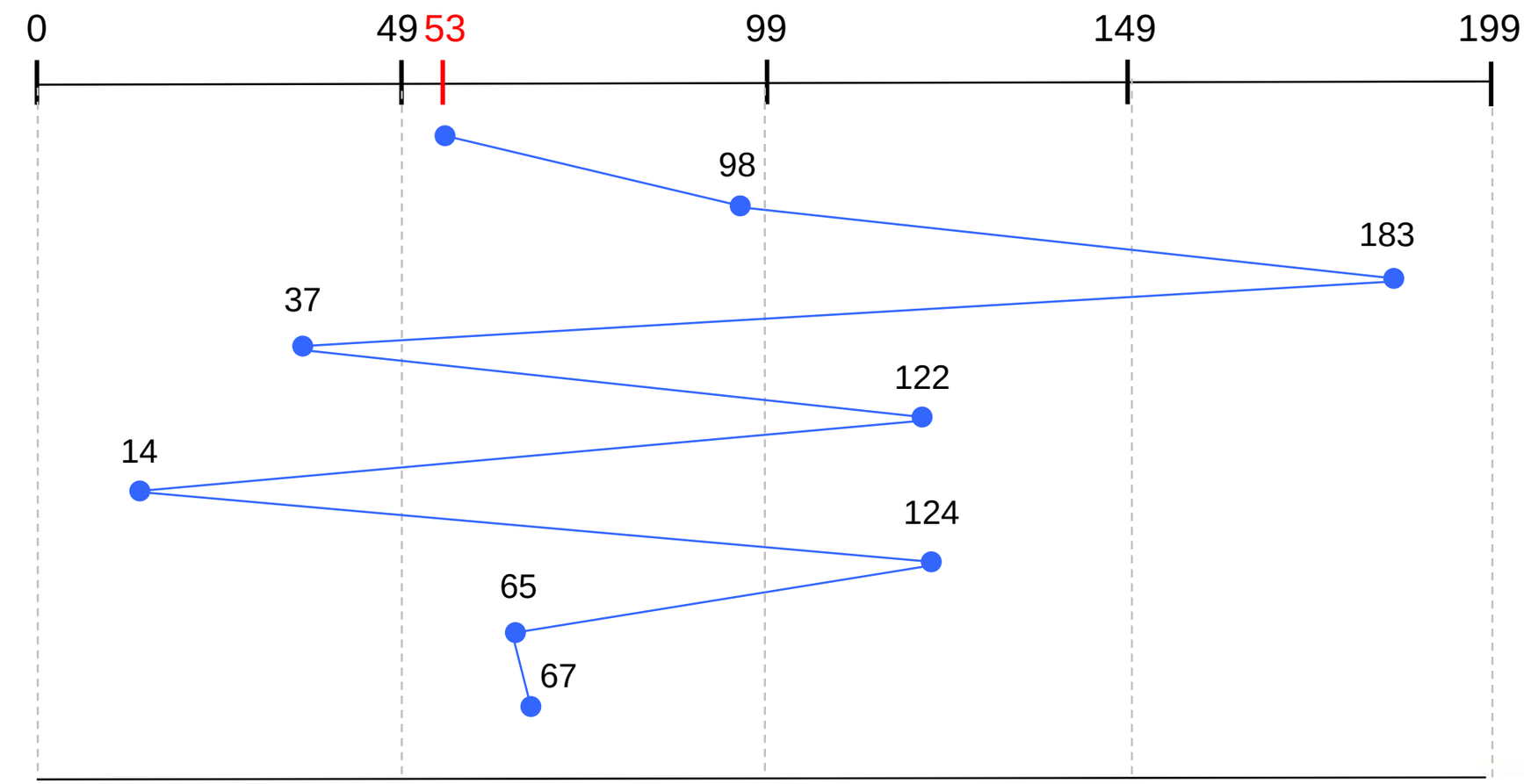
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Solution 3 :

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- A request queue :

98, 183, 37, 122, 14, 124, 65, 67



Total distance =

(98 - 53) +

(183 - 98) +

(183 - 37) +

(122 - 37) +

(122 - 14) +

(124 - 14) +

(124 - 65) +

(67 - 65)

$= 45 + 85 + 146 + 85 + 108 + 110 + 59 + 2 = 640$ cylinders.



Disk Scheduling

(b) SSTF

- This algorithm selects the request with the minimum seek time from the current head position.
- SSTF scheduling is a form of SJF scheduling; may cause starvation of some requests.
- Next example illustrates a total head movement of 236 cylinders.





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Example 4 :

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We illustrate the **SSTF** disk scheduling algorithms with the following information:

- ✓ A disk drive with 200 cylinders (0-199);
- ✓ A request queue :

  98, 183, 37, 122, 14, 124, 65, 67
- ✓ Head pointer 53.

Calculate the total distance (in cylinders) that the disk arm moves to satisfy the requests.



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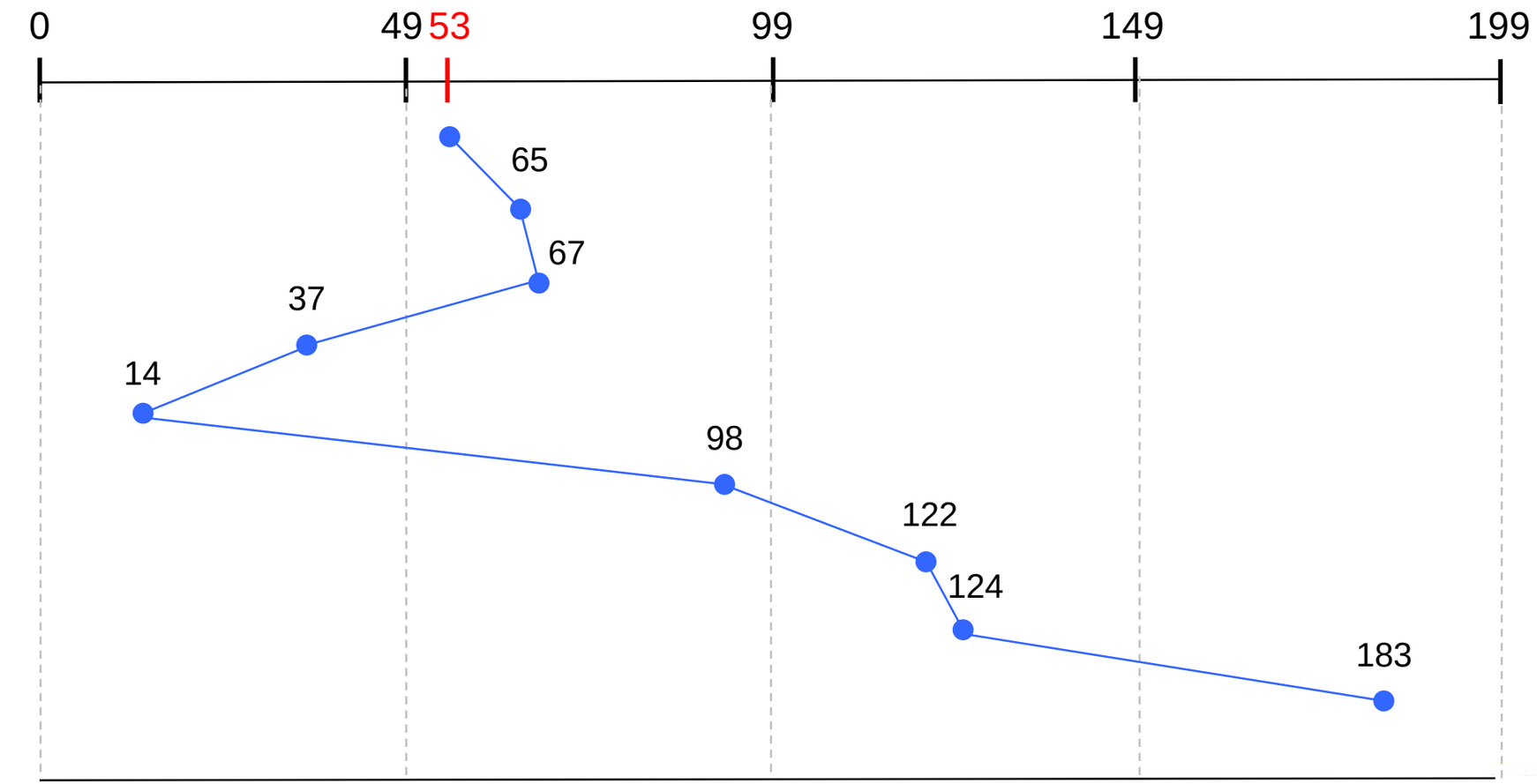
Disk Scheduling

Solution 4 :

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- A request queue :

98, 183, 37, 122, 14, 124, 65, 67



Total distance =

(65 - 53) +

(67 - 65) +

(67 - 37) +

(37 - 14) +

(98 - 14) +

(122 - 98) +

(124 - 122) +

(183 - 124)

$= 12 + 2 + 30 + 23 + 84 + 24 + 2 + 59 = 236$ cylinders (head movement).



Disk Scheduling

- Although **SSTF** algorithm is substantial improvement over **FCFS** algorithm, it is not optimal.
- We can do better by moving the head from 53 to 37, even though the latter is not closest.
- This strategy reduces the total head movement to 208 cylinders.

Exercise 12.1 : Based on previous example, redraw the entire head movement of the disk if the head moving from 53 to 37 instead of 53 to 65.



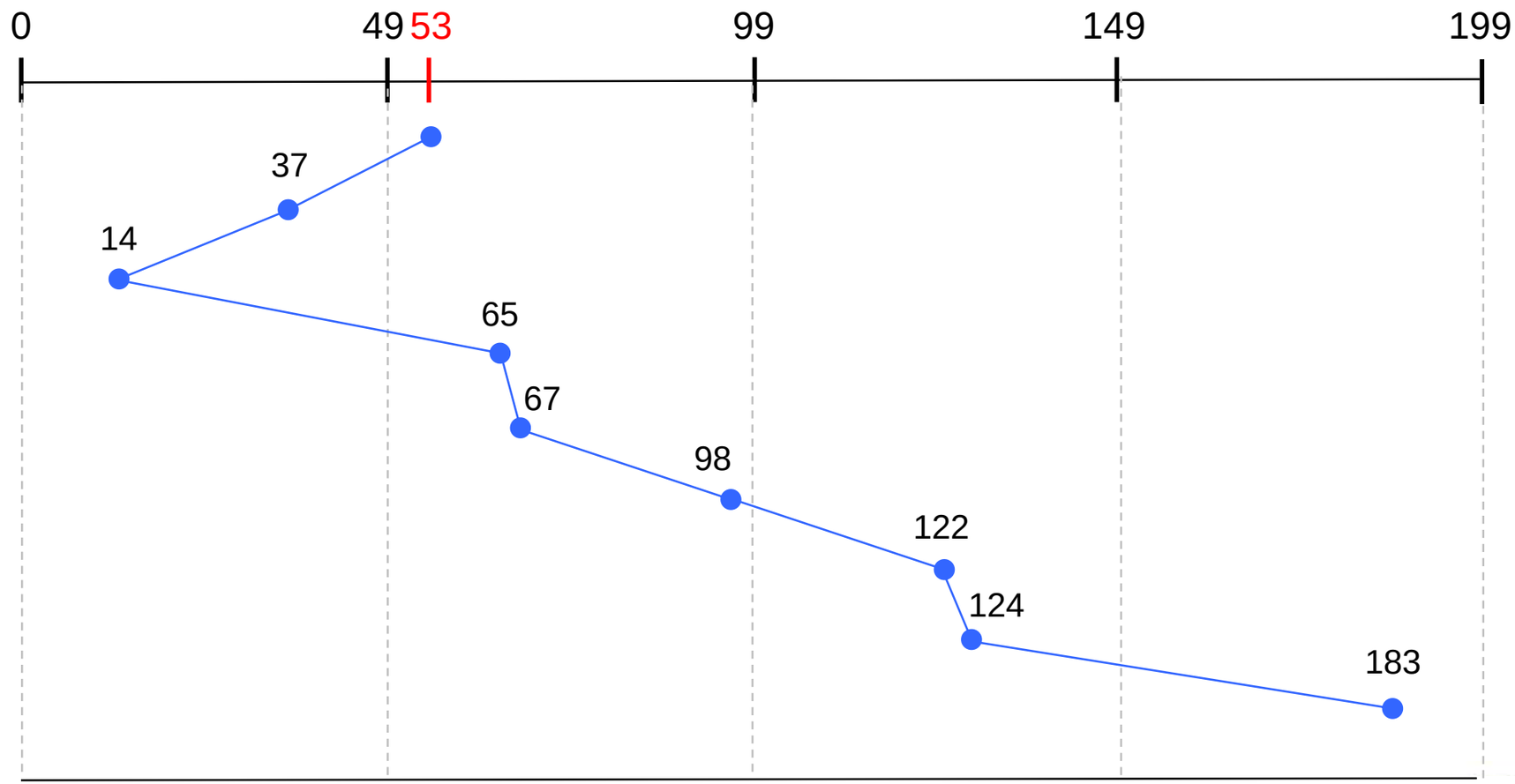
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Solution 12.1

- A request queue :

98, 183, 37, 122, 14, 124, 65, 67



Total distance =

$(53 - 37) +$

$(37 - 14) +$

$(65 - 14) +$

$(67 - 65) +$

$(98 - 67) +$

$(122 - 98) +$

$(124 - 122) +$

$(183 - 124)$

$= 16 + 23 + 51 + 2 + 31 + 24 + 2 + 59 = 208$ cylinders (head movement).



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Disk Scheduling

(c - i) SCAN

- The disk arm starts at one end of the disk, and moves **toward** the other end, servicing requests until it gets to the other end of the disk, where the head movement is **reversed** and servicing continues.
- **SCAN algorithm** sometimes called the **elevator algorithm.**

- Next example illustrates total head movement of 208 cylinders.
- But note that if requests are uniformly dense, largest density at other end of disk and those wait the longest.





Disk Scheduling

Example 5 :

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We illustrate the **SCAN** disk scheduling algorithms with the following information:

- ✓ A disk drive with 200 cylinders (0-199);
- ✓ A request queue :

  98, 183, 37, 122, 14, 124, 65, 67
- ✓ Head pointer 53.
- ✓ **Direction**: the disk arm is moving toward 0.

Calculate the total distance (in cylinders) that the disk arm moves to satisfy the requests.



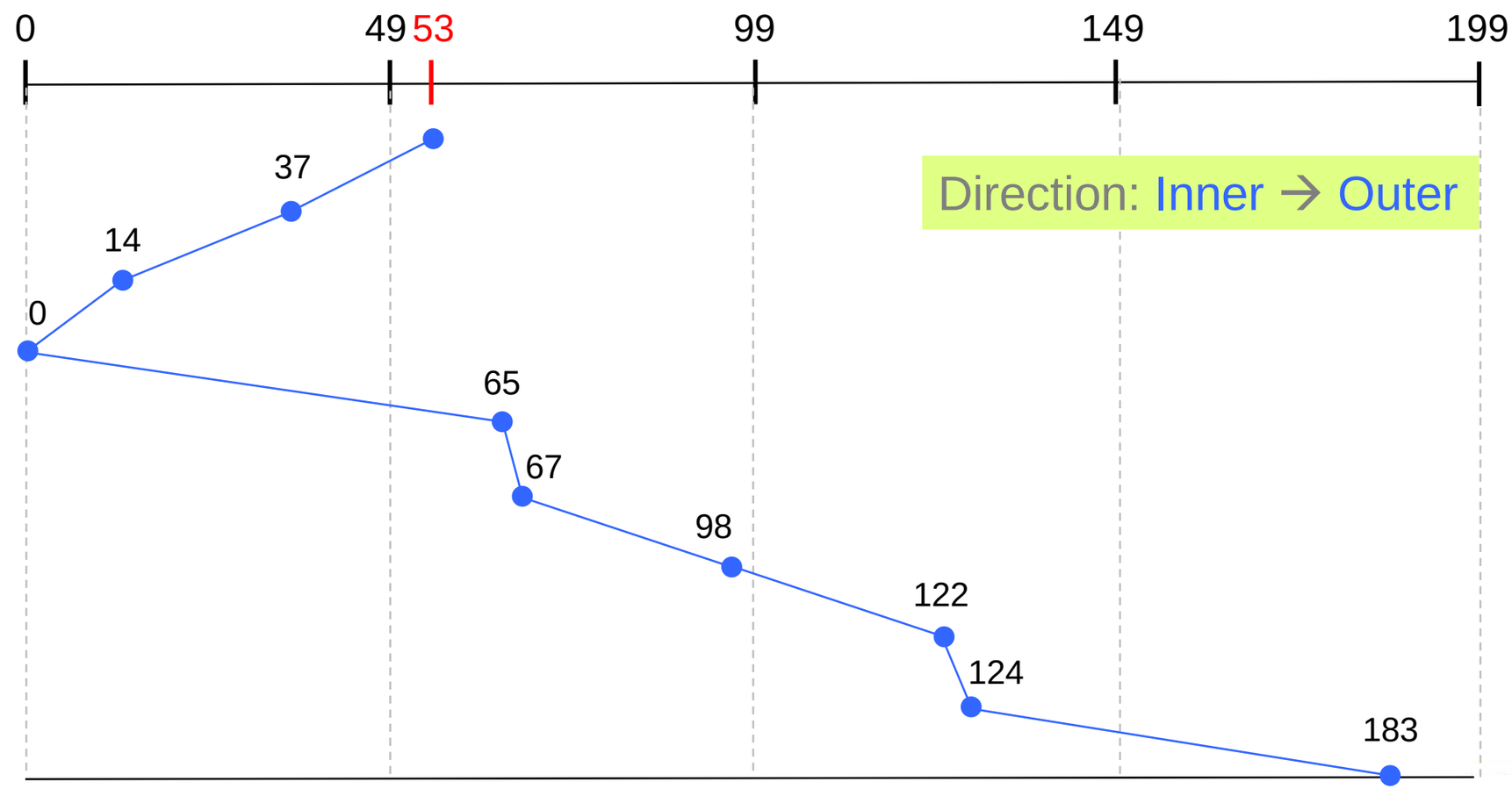
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Disk Scheduling

Solution 5 :

- A request queue :

98, 183, 37, 122, 14, 124, 65, 67



Total distance =

$(53 - 37) +$

$(37 - 14) +$

$(14 - 0) +$

$(65 - 0) +$

$(67 - 65) +$

$(98 - 67) +$

$(122 - 98) +$

$(124 - 122) +$

$(183 - 124)$

$= 25 + 23 + 14 + 65 + 2 + 31 + 24 + 2 + 59 = 245$ cylinders (head movement).



Disk Scheduling

Exercise 12.2 :

We illustrate the **SCAN** disk scheduling algorithms with the following information:

- ✓ A disk drive with 200 cylinders (0-199);
- ✓ A request queue :

  98, 183, 37, 122, 14, 124, 65, 67
- ✓ Head pointer 53.
- ✓ **Direction**: the disk arm is moving to innermost.

Calculate the total distance (in cylinders) that the disk arm moves to satisfy the requests.



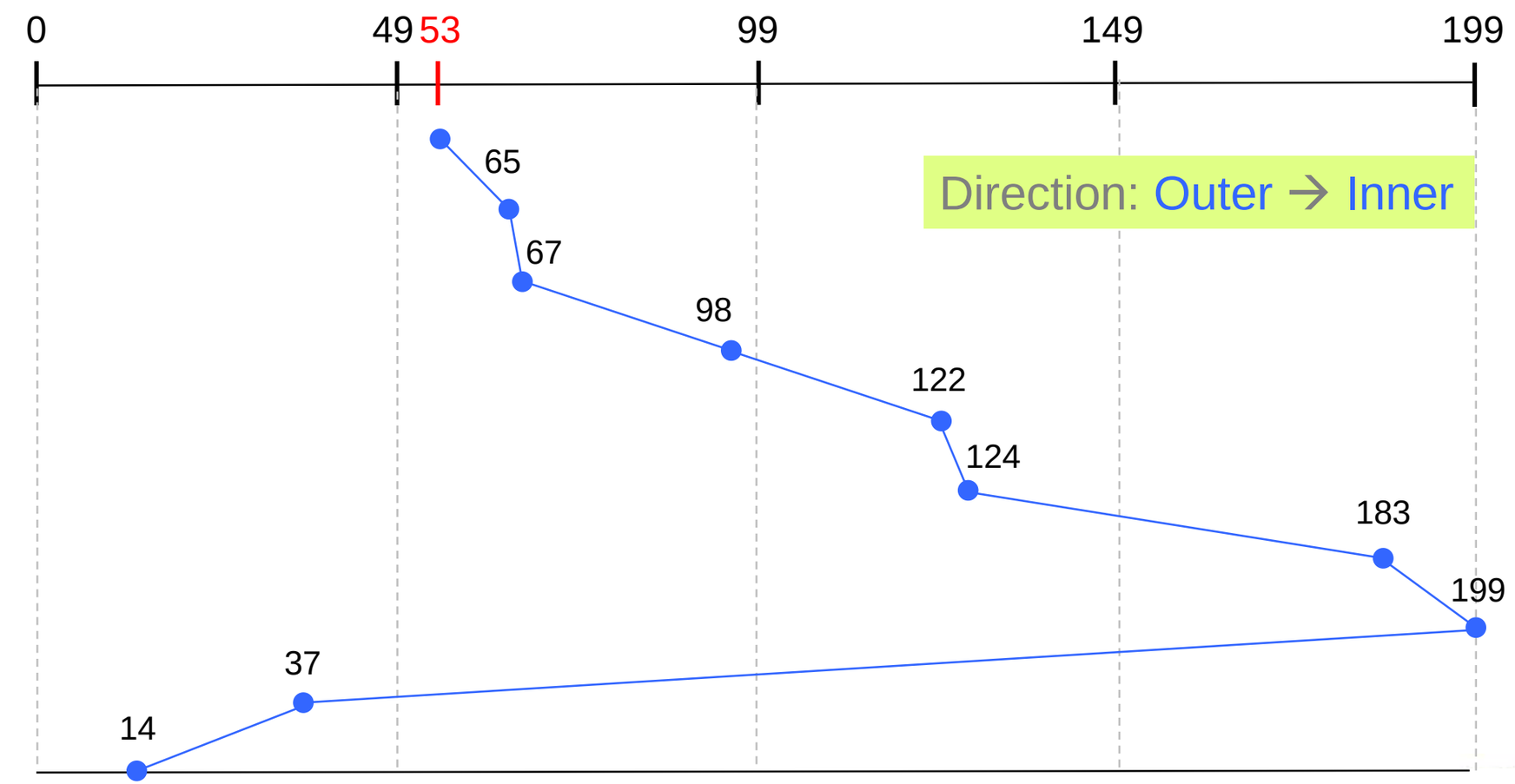
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Disk Scheduling

Solution 12.2 :

- A request queue :

98, 183, 37, 122, 14, 124, 65, 67



Total distance =

$(65 - 53) +$

$(67 - 65) +$

$(98 - 67) +$

$(122 - 98) +$

$(124 - 122) +$

$(183 - 124) +$

$(199 - 183) +$

$(199 - 37) +$

$(37 - 14)$

$= 12 + 2 + 30 + 23 + 84 + 24 + 2 + 59 = 331$ cylinders (head movement).



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Disk Scheduling

(c-ii) C-SCAN

- Provides a more uniform wait time than **SCAN**.
- The head moves from one end (**inward sweep**) of the disk to the other, servicing requests as it goes:
  - When it reaches the other end, however, it immediately **returns** to the beginning of the disk, without servicing any requests on the return trip.
- Treats the cylinders as a circular list that wraps around from the last cylinder to the first one.



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Example 6 :

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We illustrate the **C-SCAN** disk scheduling algorithms with the following information:

- ✓ A disk drive with 200 cylinders (0-199);
- ✓ A request queue :

  98, 183, 37, 122, 14, 124, 65, 67
- ✓ Head pointer 53.

Calculate the total distance (in cylinders) that the disk arm moves to satisfy the requests.



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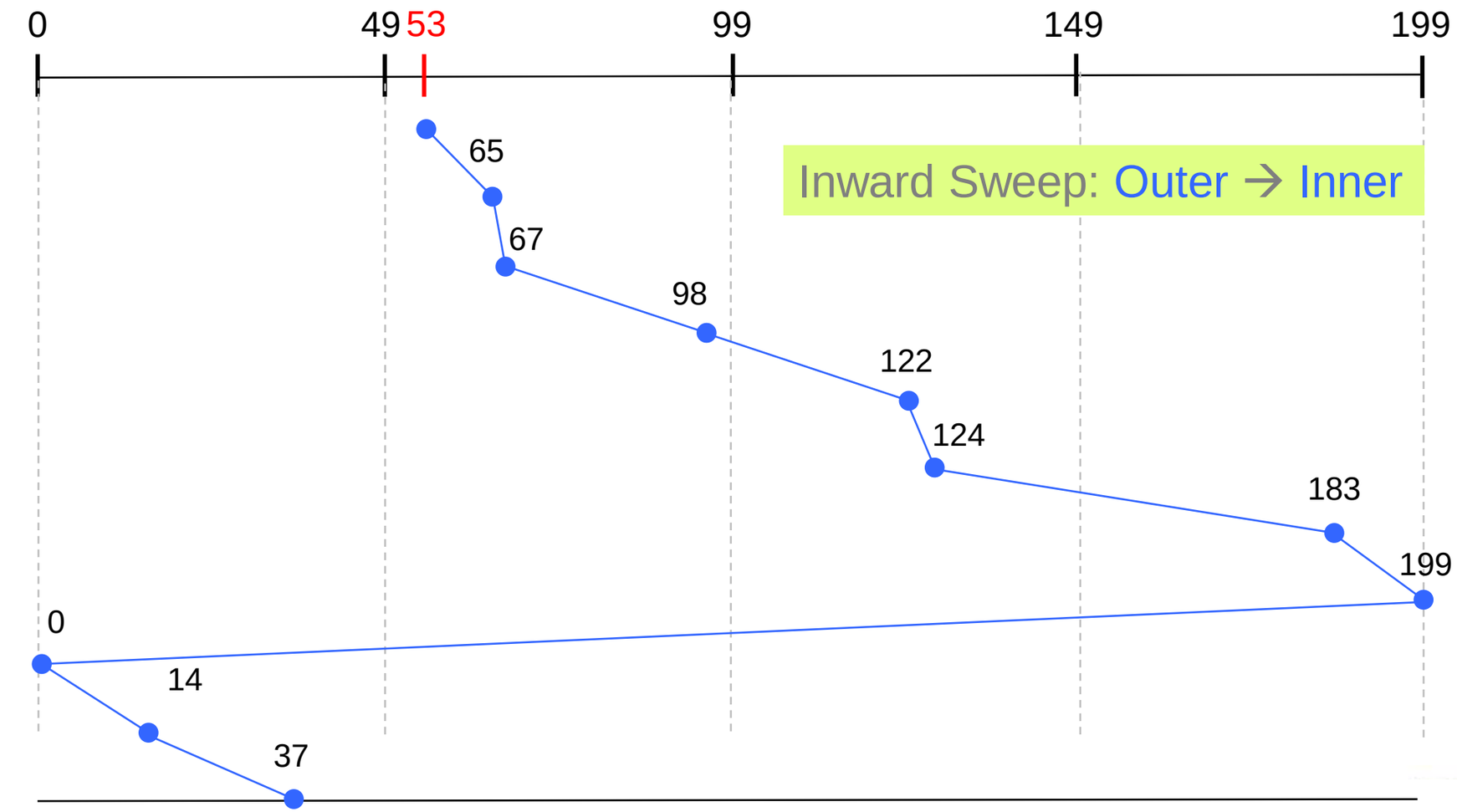
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Solution 6 :

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- A request queue :

98, 183, 37, 122, 14, 124, 65, 67



Total distance =

$(65 - 53) +$

$(67 - 65) +$

$(98 - 67) +$

$(122 - 98) +$

$(124 - 122) +$

$(183 - 124) +$

$(199 - 183) +$

$(199 - 0) +$

$(14 - 0) +$

$(37 - 14)$

$= 12 + 2 + 31 + 24 + 2 + 59 + 16 + 199 + 14 + 23 = 382$ cylinders (head movement).



Disk Scheduling

(d) LOOK and C-LOOK

- As described before, both SCAN and C-SCAN move the disk arm across the full width of the disk.
- The arms goes only as far as the final request in each direction, then reverses direction immediately, without going all the way to the end of the disk.
- Version of SCAN and C-SCAN that follow this pattern is called **LOOK** and **C-LOOK** disk scheduling.





Disk Scheduling

Example 7 :

We illustrate the **LOOK** disk scheduling algorithms with the following information:

- ✓ A disk drive with 200 cylinders (0-199);
- ✓ A request queue :

  98, 183, 37, 122, 14, 124, 65, 67
- ✓ Head pointer 53.
- ✓ **Direction**: the disk arm is moving toward 0.

Calculate the total distance (in cylinders) that the disk arm moves to satisfy the requests.



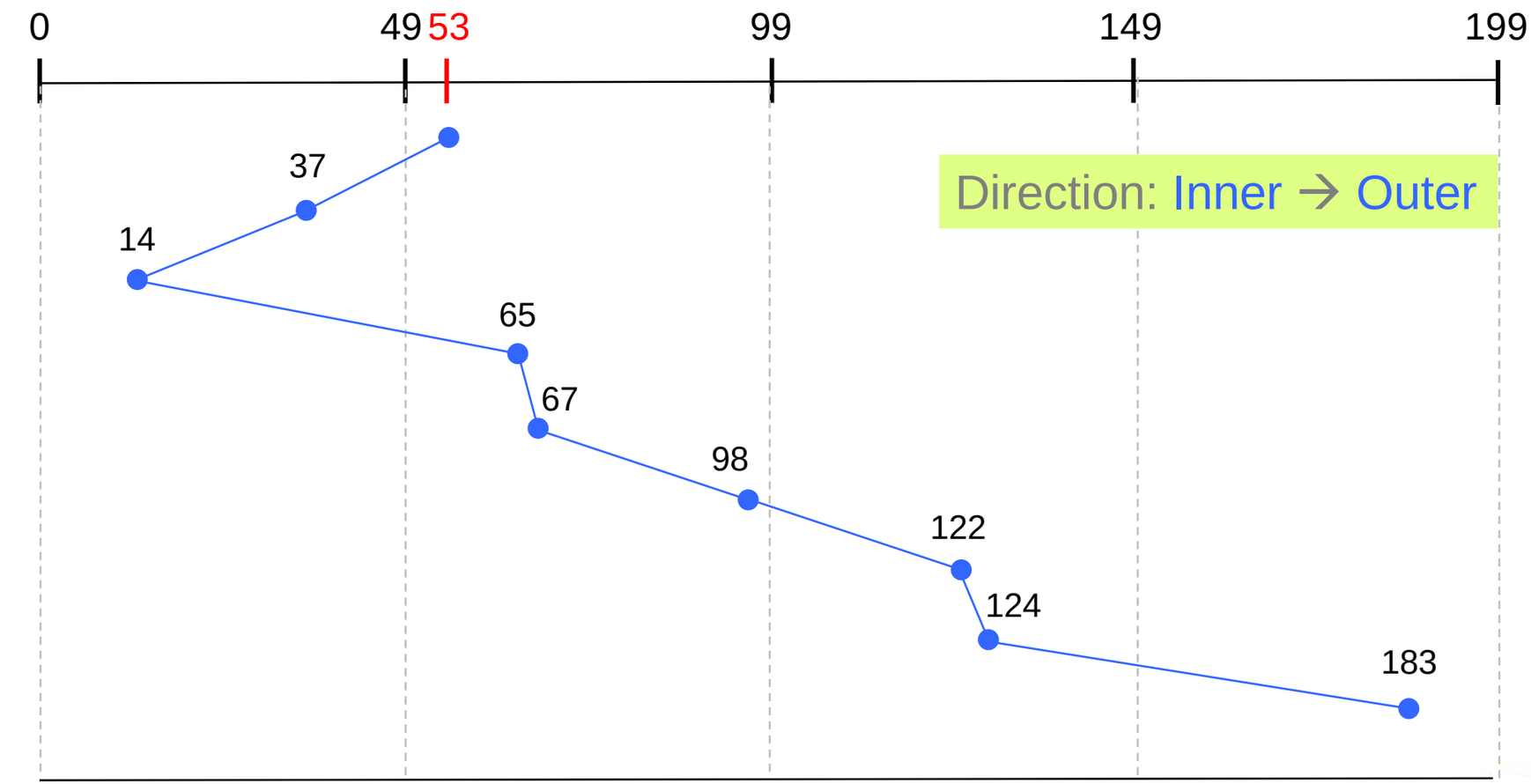
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Disk Scheduling

Solution 7 :

- A request queue :

98, 183, 37, 122, 14, 124, 65, 67



Total distance =

$(53 - 37) +$

$(37 - 14) +$

$(65 - 14) +$

$(67 - 65) +$

$(98 - 67) +$

$(122 - 98) +$

$(124 - 122) +$

$(183 - 124)$

$= 16 + 23 + 51 + 2 + 31 + 24 + 2 + 59 = 208$ cylinders (head movement).



Disk Scheduling

Exercise 12.3 :

We illustrate the **LOOK** disk scheduling algorithms with the following information:

- ✓ A disk drive with 200 cylinders (0-199);
- ✓ A request queue :

  98, 183, 37, 122, 14, 124, 65, 67
- ✓ Head pointer 53.
- ✓ **Direction**: the disk arm is moving to innermost.

Calculate the total distance (in cylinders) that the disk arm moves to satisfy the requests.

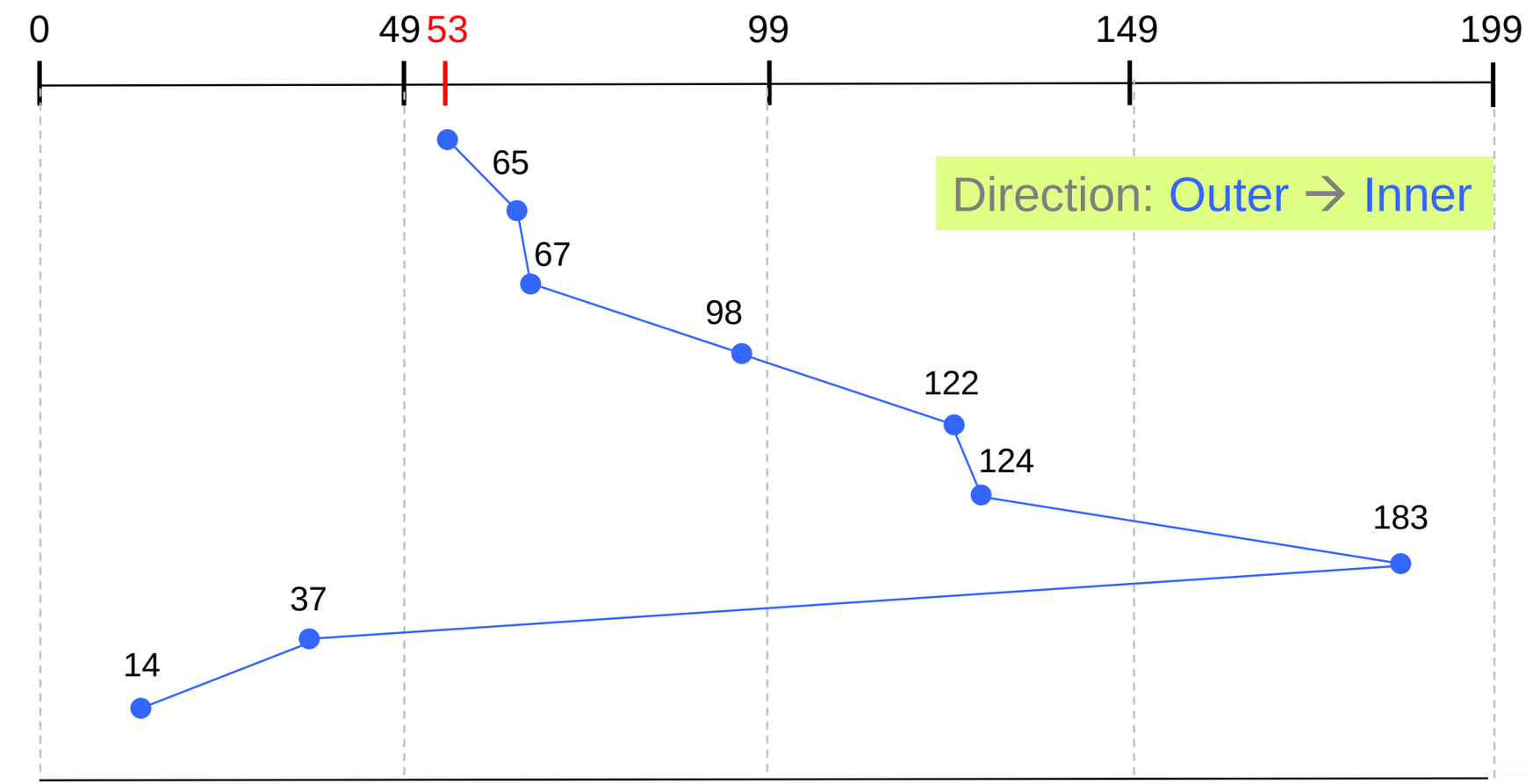


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Solution 12.3 :

- A request queue :

98, 183, 37, 122, 14, 124, 65, 67



Total distance =

$(65 - 53) +$
$(67 - 65) +$
$(98 - 67) +$
$(122 - 98) +$
$(124 - 122) +$
$(183 - 124) +$
$(183 - 37) +$
$(37 - 14)$

$= 12 + 2 + 30 + 23 + 84 + 24 + 2 + 59 = 331$ cylinders (head movement).



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Example 8 :

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We illustrate the **C-LOOK** disk scheduling algorithms with the following information:

- ✓ A disk drive with 200 cylinders (0-199);
- ✓ A request queue :

  98, 183, 37, 122, 14, 124, 65, 67
- ✓ Head pointer 53.

Calculate the total distance (in cylinders) that the disk arm moves to satisfy the requests.



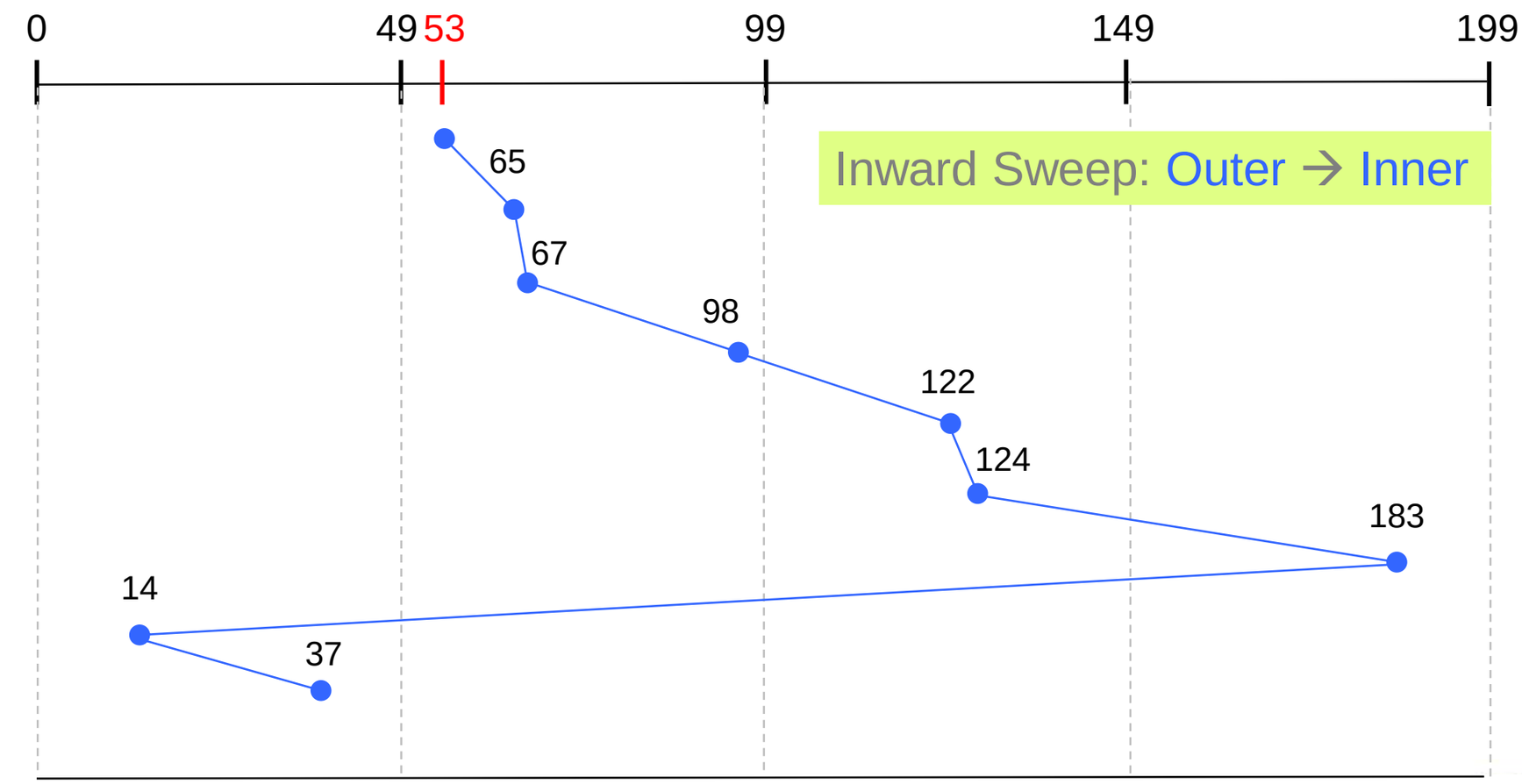
Disk Scheduling

Solution 8 :

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- ✓ A request queue :

98, 183, 37, 122, 14, 124, 65, 67



Total distance =

$(65 - 53) +$

$(67 - 65) +$

$(98 - 67) +$

$(122 - 98) +$

$(124 - 122) +$

$(183 - 124) +$

$(183 - 14) +$

$(37 - 14)$

$= 12 + 2 + 31 + 24 + 2 + 59 + 169 + 23 = 322$ cylinders (head movement).